

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

INFORMATION

The specified values and value ranges correspond to the OK state in each case.

Line test without connected components

Short circuit to B+:

♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Passenger's side front

⚠ WARNING

Uncontrolled triggering of the airbag

- Contusions, crushing injuries or death

- ▶ **Never hang the PIWIS Tester** on or over the steering wheel.
- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
- ▶ Wait for at least one minute before starting work on the airbag system.
- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- Contusions, crushing injuries or death
- Traffic hazard

- ▶ Never drive with special tools and/ or commercially available tools attached instead of the airbag units.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- **Supply voltage** > 9.1 V
- Ignition on > 5 Second(s), then:
- Ignition off > 2 Second(s), then:
- Ignition on > 5 Second(s)
- Airbag triggering unit Internal function test

Possible fault causes

- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
- ♦ Electrical fault in circuit - Electrical connection between two ignition circuits (mutual short circuit). Short circuit in wiring harness of two ignition circuits
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- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
- ♦ Electrical fault in circuit - Electrical connection between two ignition circuits (mutual short circuit). Short circuit in wiring harness of two ignition circuits
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Airbag sensor** Passenger's side front incorrect
- ♦ **Airbag sensor** Passenger's side front Faulty

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

INFORMATION

The specified values and value ranges correspond to the OK state in each case.

Line test without connected components

Short circuit to B+:

♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

⚠ WARNING

Uncontrolled triggering of the airbag

- Contusions, crushing injuries or death

- ▶ **Never hang the PIWIS Tester** on or over the steering wheel.
- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
- ▶ Wait for at least one minute before starting work on the airbag system.
- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- Contusions, crushing injuries or death
- Traffic hazard

- ▶ Never drive with special tools and/ or commercially available tools attached instead of the airbag units.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- **Supply voltage** > 9.1 V
- Ignition on > 5 Second(s), then:
- Ignition off > 2 Second(s), then:
- Ignition on > 5 Second(s)
- Airbag triggering unit Internal function test

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- ♦ Electrical fault in circuit - Electrical connection between two ignition circuits (mutual short circuit). Short circuit in wiring harness of two ignition circuits
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Airbag sensor** Driver's side rear incorrect
- ♦ **Airbag sensor** Driver's side rear Faulty

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

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- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

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- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

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- ♦ Setpoint resistance **> 0.9 MOhm**

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Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

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Uncontrolled triggering of the airbag

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Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

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Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

⚠ WARNING

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Line test without connected components

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Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

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Fault effects

- ♦ Warning light for Airbag is **ON**
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Diagnosis information - Airbag

Airbag sensor Driver's side rear

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- ♦ Warning light for Airbag is **ON**
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Diagnosis information - Airbag

Airbag sensor Driver's side rear

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Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

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INFORMATION

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Short circuit between lines :

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Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Driver's side rear

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Short circuit between lines :

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Fault effects

- ♦ Warning light for Airbag is **ON**
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Diagnosis information - Airbag

Airbag sensor Passenger's side rear

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